Discrete - 1.2.18

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Question

The z-transform $\mathbf{X[z]}$ of a sequence x[n] is given by $\mathbf{X[z]} = \frac{0.5}{1-2z^{-1}}$. It is given that the region of convergence of $\mathbf{X[z]}$ includes the unit circle. The value of x[0] is:

1. -0.5
2. 0
3. 0.25
4. 0.5

(EC 2007)

Solution

The Z-transform of a discrete-time sequence x[n] is defined as :

$$\mathbf{X}(\mathbf{z}) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} \tag{1}$$

The ROC (region of covergence) is the set of values of z in the complex plane for which the Z-transform sum converges absolutely (i.e., gives a finite value)

A pole of a Z-transform is defined as the value of z that makes the X[z] to blow up to infinity (i.e., denominator of X[z] equal to zero)

Solution

Set :

$$1 - 2z^{-1} = 0$$
$$\Rightarrow 2z^{-1} = 1$$
$$\Rightarrow z^{-1} = \frac{1}{2}$$

$$\Rightarrow z = 2 \tag{2}$$

Therefore , there is a single pole at z = 2, which lies at a radius of $|z| = 2$ in the complex plane.

Solution

The pole of $X(z)$ is at $z = 2$, so two possible ROCs exist:

$$\text{ROC}_1 : |z| > 2 \quad \text{(right-sided, causal)} \tag{3}$$
$$\text{ROC}_2 : |z| < 2 \quad \text{(left-sided, anti-causal)} \tag{4}$$

Since the unit circle $|z| = 1$ must lie inside the ROC, and:

$$|z| = 1 < 2 \tag{5}$$

The unit circle lies inside $|z| < 2$, NOT inside $|z| > 2$.

ROC : $|z| < 2$

$$\text{This means } x[n] \text{ is a } \textbf{left-side(anti-causal)} \text{ sequence.} \tag{6}$$

Solution

Given Expression ,

$$\mathbf{X}[\mathbf{z}] = \frac{0.5}{1 - 2z^{-1}} \tag{7}$$

by multiplying numerator and denominator with z

$$\mathbf{X}[\mathbf{z}] = \frac{0.5z}{z - 2} \tag{8}$$

$$\mathbf{X}[\mathbf{z}] = -0.5\left(\frac{z}{2}\right)\left(\frac{1}{1 - \frac{z}{2}}\right) \tag{9}$$

Solution

Expanding $\left(\frac{1}{1-\frac{z}{2}}\right)$ using Geometric Series

$$\Rightarrow \frac{1}{1-\frac{z}{2}} = \sum_{k=0}^{\infty}\left(\frac{z}{2}\right)^k \tag{10}$$

(Since , $|z| < 2 \Rightarrow |\frac{z}{2}| < 1$)

$$\mathbf{X}(\mathbf{z}) = -0.5\frac{z}{2}\sum_{k=0}^{\infty}\left(\frac{z}{2}\right)^k \tag{11}$$

$$= -0.5\sum_{k=0}^{\infty}\frac{z^{k+1}}{2^{k+1}} \tag{12}$$

by the definition of Z-transform

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} \tag{13}$$

Solution

On comparing powers

$$z^{k+1} = z^{-n} \Rightarrow n = -(k+1) \tag{14}$$

$$\text{Hence, } k = -n - 1 \tag{15}$$

$$x[n] = -0.5 \cdot 2^n, \quad \text{for } n \leq -1 \tag{16}$$

$$\therefore \quad x[n] = -0.5 \cdot 2^n \, u[-n-1] \tag{17}$$

where $u[-n-1]$ is the unit step function defined as:

$$u[-n-1] = \begin{cases} 1 & n \leq -1 \\ 0 & \text{otherwise} \end{cases} \tag{18}$$

Solution

$$x[0] = -0.5\left(2^0\right)(u[-0-1]) \tag{19}$$
$$= -0.5\,(1)\,(u[-1]) \tag{20}$$
$$= -0.5\,(1)\,(0) = 0 \tag{21}$$
$$\text{Therefore, } x[0] = 0 \text{ (Option 2)} \tag{22}$$

Solution - Method-2

We have :

$$\mathbf{X}[\mathbf{z}] = \frac{0.5}{1 - 2z^{-1}} \tag{23}$$

On Rearranging :

$$\mathbf{X}[\mathbf{z}]\left(1 - 2z^{-1}\right) = 0.5 \tag{24}$$

Applying Inverse Z-Transform on both sides:

$$\mathbf{X}(\mathbf{z}) - 2z^{-1}\mathbf{X}(\mathbf{z}) = 0.5 \tag{25}$$

Using the following Z-Transform properties:

$$\mathbf{X}(\mathbf{z}) \longleftrightarrow x(n) \tag{26}$$

$$z^{-1}\mathbf{X}(\mathbf{z}) \longleftrightarrow x(n-1) \quad \text{(Time Delay Property)} \tag{27}$$

$$0.5 \longleftrightarrow 0.5\delta(n) \quad \text{(Inverse Z-transform of constant)} \tag{28}$$

Applying Inverse Z-Transform directly:

$$\mathcal{Z}^{-1}\left\{X(z) - 2z^{-1}X(z)\right\} = \mathcal{Z}^{-1}\{0.5\} \tag{29}$$

Solution (Method-2)

The difference equation is:

$$\boxed{x(n) - 2x(n-1) = 0.5\delta(n)} \tag{30}$$

$$\text{Given difference equation:} \quad x[n] - 2x[n-1] = 0.5\,\delta[n] \tag{31}$$

$$\text{Since ROC is } |z| < 2, \text{ the sequence is left-sided.} \tag{32}$$

$$\Rightarrow x[n] = 0 \quad \text{for } n \geq 0 \tag{33}$$

$$\Rightarrow x[0] = 0 \tag{34}$$

$$\text{At } n = 0: \quad x[0] - 2x[-1] = 0.5 \tag{35}$$

$$0 - 2x[-1] = 0.5 \tag{36}$$

$$x[-1] = -0.25 \tag{37}$$

Solution

$$\text{For } n \neq 0,\ \delta[n] = 0: \quad x[n] - 2x[n-1] = 0 \tag{38}$$

$$\Rightarrow x[n] = 2x[n-1] \tag{39}$$

$$\Rightarrow x[n-1] = \frac{x[n]}{2} \tag{40}$$

$$\text{Using backward recursion:} \tag{41}$$

$$x[-2] = \frac{x[-1]}{2} = -0.125 \tag{42}$$

$$x[-3] = \frac{x[-2]}{2} = -0.0625 \tag{43}$$

$$\vdots \tag{44}$$

$$\text{Thus, the general form is:} \quad x[n] = -0.5 \cdot 2^n \quad \text{for } n \leq -1 \tag{45}$$

Solution

If we plot the respective x[n] values , we will get the plot in slide 20 .

Python code

```python
import numpy as np
import matplotlib.pyplot as plt
from scipy import signal

# Given
num = [0.5]
den = [1, -2]

# Poles
_, poles, _ = signal.tf2zpk(num, den)

# Left-sided inverse Z-transform
n_vals = np.arange(-5, 6)
x_n = np.array([-0.5 * (2.0**n) if n <= -1 else 0.0 for n in
    n_vals])

print(f"x[0] = {x_n[n_vals == 0][0]}")
```

Python code

```
# plotting
fig, axes = plt.subplots(1, 2, figsize=(12, 5))
fig.suptitle(r"Z-Transform: $X(z) = \frac{0.5}{1-2z^{-1}}$, ROC:
    $|z|<2$ (Left-sided)",
fontsize=13, fontweight='bold')

ax1 = axes[0]
ax1.stem(n_vals, x_n, basefmt='k-', linefmt='steelblue',
    markerfmt='bo')
ax1.set_title(r"x[n] vs n")
ax1.set_xlabel("n")
ax1.set_ylabel("x[n]")
ax1.grid(True)

ax2 = axes[1]
theta = np.linspace(0, 2*np.pi, 300)
roc_circle = plt.Circle((0, 0), abs(poles[0]), alpha=0.15,label='
    ROC: |z| < 2')
```

Python code

```
ax2.add_patch(roc_circle)
ax2.plot(np.cos(theta), np.sin(theta), linestyle=':', linewidth
    =1.5,label='Unit Circle')

ax2.scatter(np.real(poles), np.imag(poles), s=120,label='Pole at
    z=2')

ax2.legend()
ax2.axhline(0, linewidth=0.5)
ax2.axvline(0, linewidth=0.5)
ax2.set_title("Pole-Zero Plot")
ax2.set_xlabel("Real")
ax2.set_ylabel("Imaginary")
ax2.grid(True)
ax2.set_aspect('equal')
ax2.set_xlim(-3, 3)
ax2.set_ylim(-3, 3)
plt.tight_layout()
plt.show()
```

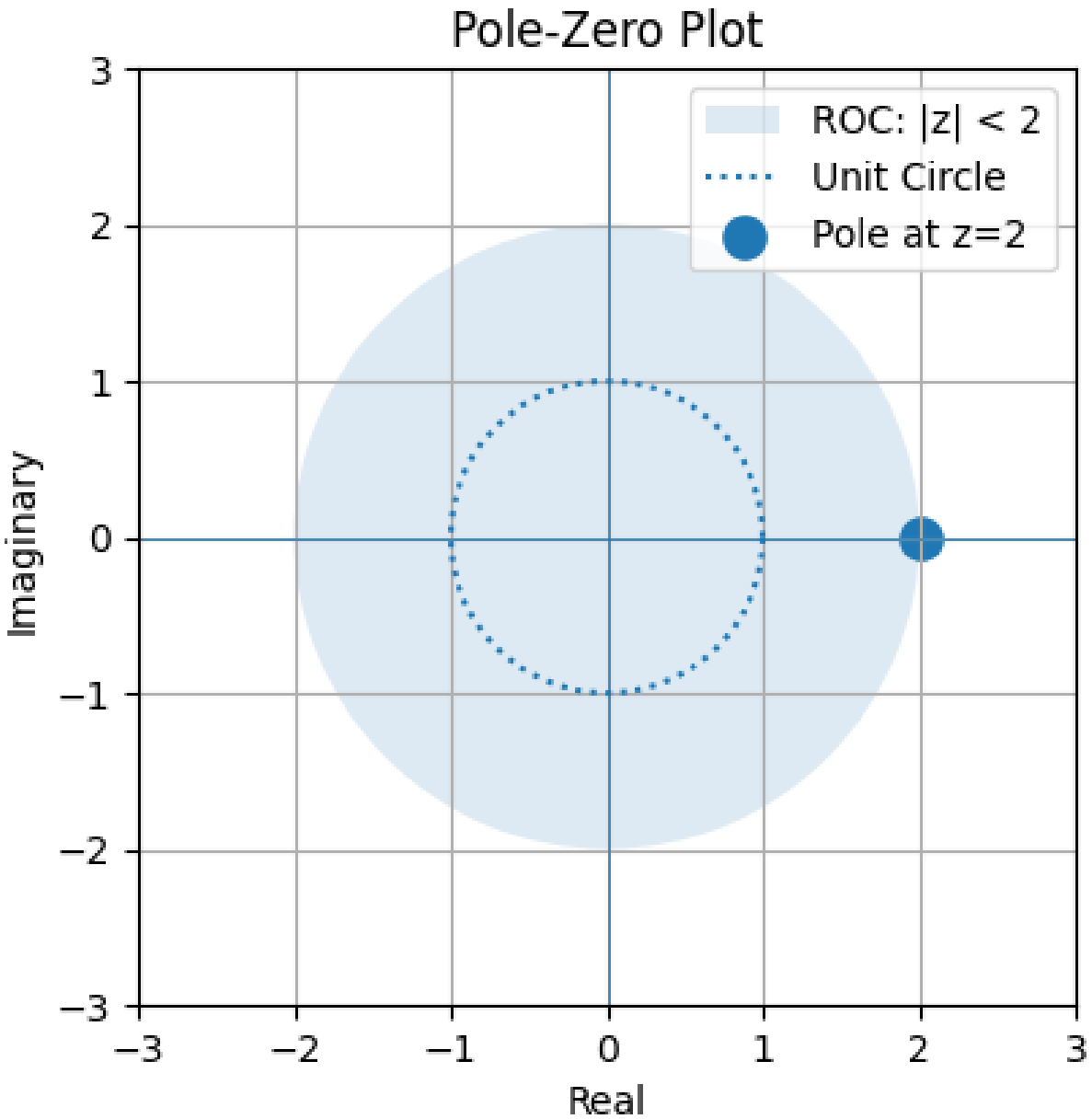
Python code (for plotting x[n] obtained from difference equation)

```
import numpy as np
import matplotlib.pyplot as plt

# x[n]
n = np.arange(-10, 11)
x = np.zeros_like(n, float)

i0 = np.where(n == 0)[0][0]
x[i0-1] = -0.25

for i in range(i0-1, 0, -1):
    x[i-1] = x[i] / 2

#plotting
plt.figure(figsize=(10, 5))
markerline, stemlines, baseline = plt.stem(n, x, basefmt='k-')
plt.setp(markerline, markersize=6)
```

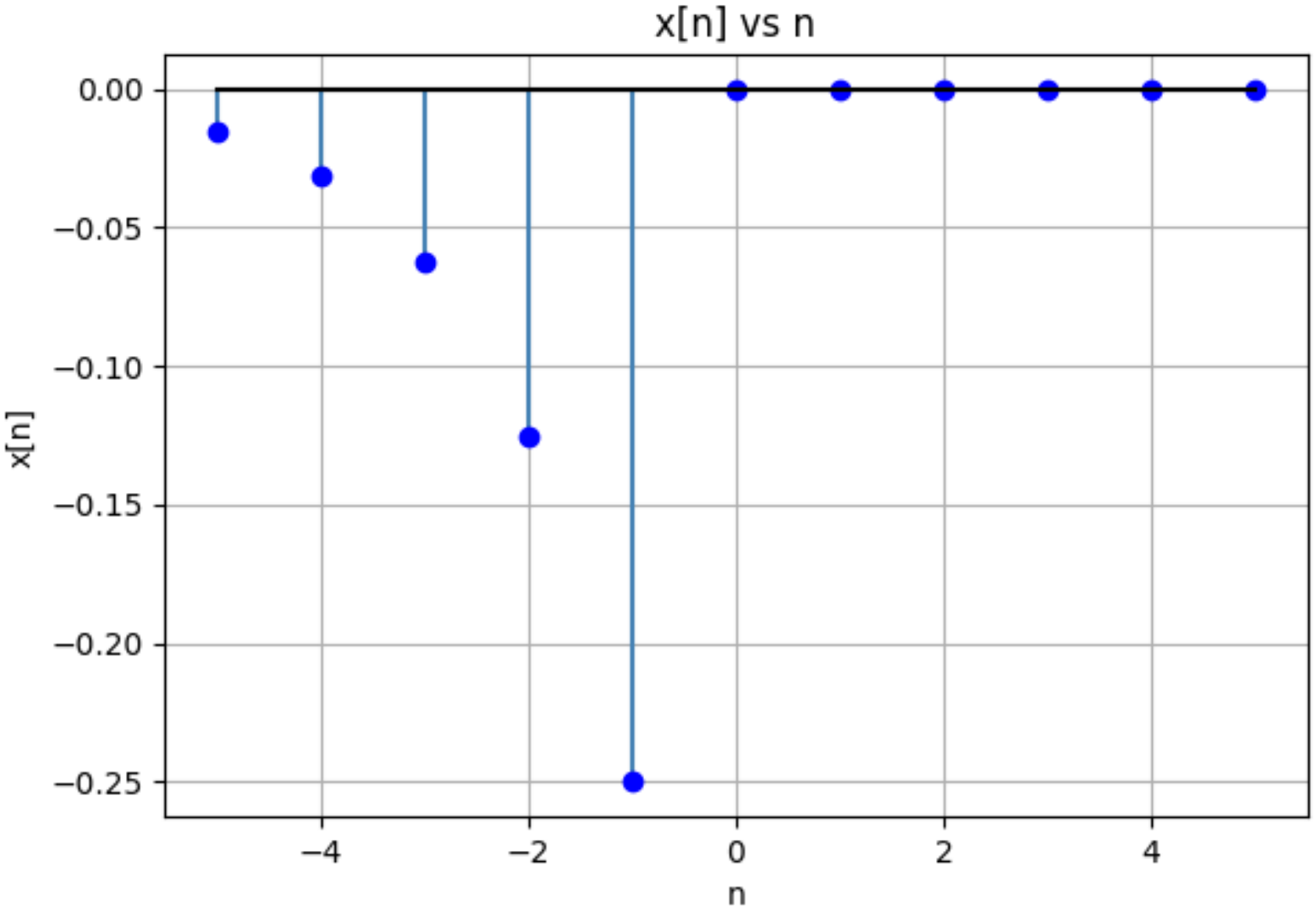
Python code(for plotting x[n] obtained from difference equation)

```
    plt.setp(stemlines, linewidth=1.5)
plt.title(" x[n] vs n (through difference equation", fontsize=13,
     fontweight='bold')
plt.xlabel("n", fontsize=11)
plt.ylabel("x[n]", fontsize=11)
plt.grid(True, alpha=0.3)
plt.xlim(-10, 10)
plt.ylim(min(x) - 0.1, 0.1)
plt.tight_layout()
plt.show()
```

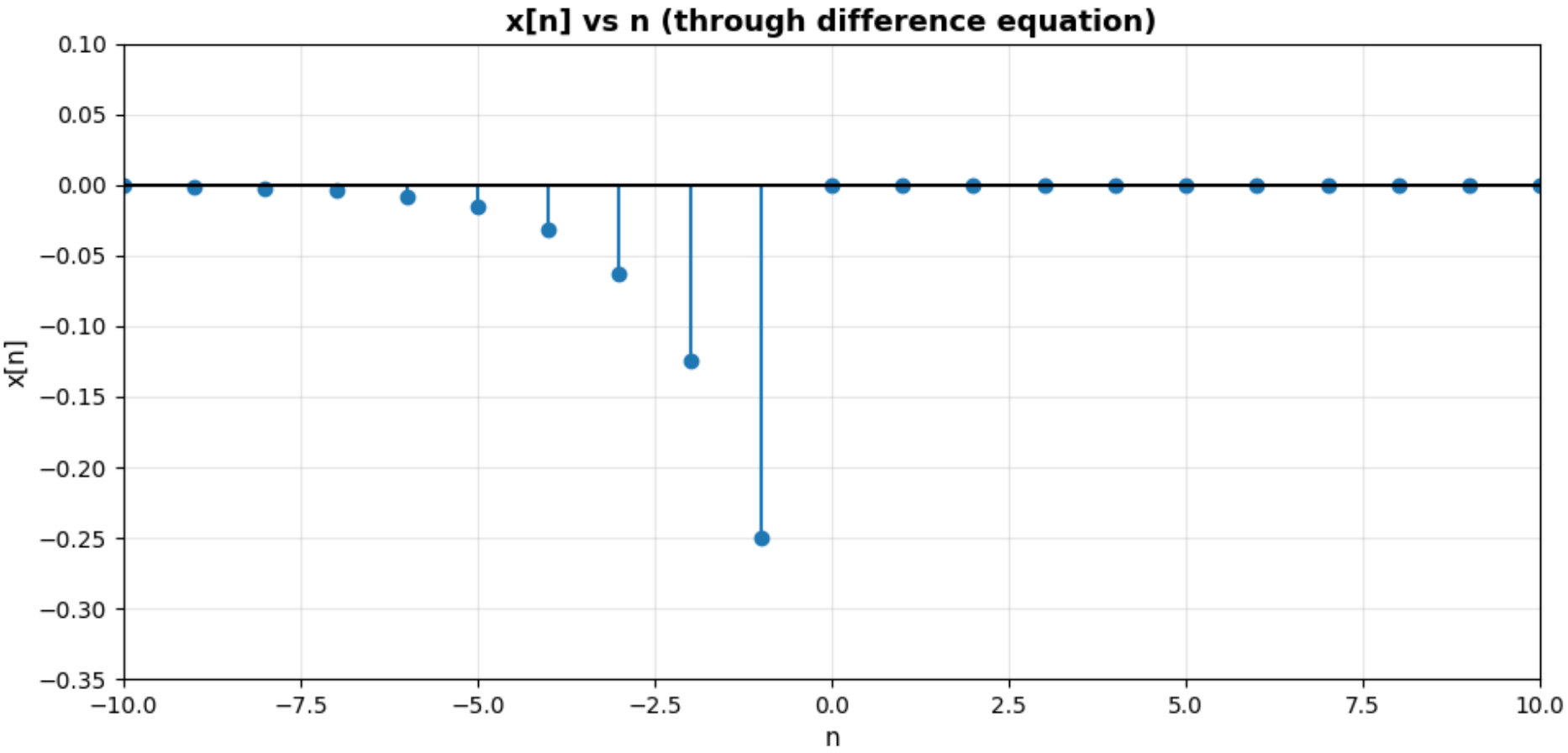
Plot


Pole-Zero Plot
ROC: |z| < 2
Unit Circle
Pole at z=2
Imaginary
Real
3
2
1
0
−1
−2
−3
−3
−2
−1
0
1
2
3

Plot



Plot


x[n] vs n (through difference equation)
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
−0.25
−0.30
−0.35
x[n]
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
n